

15. Aug 2026

Dear Mikkel,

I hope this letter finds you well. I was recently reminded of our early fieldwork days, when long mornings by the lakeside were often cold but always full of possibility. That memory returns now as I write to you with great anticipation about an exciting opportunity.

As you know, my passion has always been to unravel the mysteries of ancient communities, and today we are on the verge of a project that could deepen our understanding of daily life in early Ireland. Our interdisciplinary team of archaeologists and environmental scientists has begun investigating the crannog at Lake Catrine. These remarkable artificial islands, built from wood, brushwood, and stone, dot the Irish landscape and have long intrigued scholars. Yet their exact function—settlement, refuge, or ceremonial site—remains uncertain.

This time, however, we intend to try something new. Alongside traditional excavation, we propose to apply ancient environmental DNA (eDNA) analysis to lake sediment cores sampled next to the crannog. By sequencing genetic fragments preserved in these laminated layers, we may detect traces of crops such as barley, domestic animals like cattle or pigs, wild plants and fish, and even pathogens. In short, we could reconstruct a detailed picture of the ecosystem—and perhaps the human activity—that surrounded the site across centuries.

Your pioneering expertise in sedimentary ancient DNA is, of course, invaluable here. You know well both the remarkable insights and the challenges—contamination, degradation, interpretation—that this method entails. Still, if successful, such data could offer unprecedented evidence about how crannogs were used. Moreover, it could allow us to compare Lake Catrine to other sites across Ireland and Scotland, asking whether these structures played similar roles or served quite different purposes.

Our budget is modest, at €20,000, which we expect to cover laboratory analyses, training opportunities for students, and field logistics. I was especially thinking of your new students—might this not make an ideal project for them, combining archaeology, genetics, and environmental reconstruction?

Enclosed you will find an aerial photograph of Lake Catrine, with our coring sites marked in the upper right corner. Where would you place additional cores? I also include a photograph of the sediment sequence we retrieved: finely laminated layers, beautifully preserved. They suggest seasonal deposition and a high-resolution environmental archive, though I assure you the early morning work to get them was chilly indeed!

I very much hope you will consider joining this collaboration. Together, we could not only shed new light on this crannog but also reshape how we think about such enigmatic sites across Ireland.

With great anticipation,

Your old friend,

Prof. Eoghan Ó Muircheartaigh



Here is a aerial photo of the lake and area around it. With the coring sites in the upper right corner.

Below here is a picture of the lake sediment record, we found beautiful lamination of layers. It seems that the fieldwork was worth it, although it was cold that morning.

